Killing two birds with one stone: Evaluate sprinting horizontal force production capacities to improve performance and injury risk management.

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How the evaluation of sprinting horizontal force production capacities can macroscopically evaluate lower limb function.

Keywords: sprint mechanics; lower limb function; sports injury prevention.

Sprinting is important for athletes, coaches, and health professionals for both performance and injury risk management [1]. Sprinting can be defined as running with the intention of maximal acceleration and/or velocity. The performance of sprinting acceleration is associated with the ability to produce and apply high levels of force in the horizontal direction [2]. Such information is of great importance when we want to evaluate, monitor and/or train for sprinting. Evaluating and monitoring sprinting horizontal force production capacities is now used in field settings for sprinting performance enhancement, [3,4] as well as in more analytical evaluations such as isolated single joint strength assessments. This ability to produce horizontal force during sprinting is mainly associated with the lower limb posterior chain muscles actions [5,6]. Thus, we can consider that sprinting horizontal force production capacities evaluation allows an overall evaluation of the lower limb function during sprinting action. More specifically, this evaluation targets the overall posterior chain muscles function during sprinting action. Recent studies also supported the interest of evaluating sprinting horizontal force production for hamstring injury management [7–11].

This blog will explain how to assess sprinting horizontal force production capacities and why they can be of interest for both injury risk management and to performance improvement.

What are sprinting "horizontal force" production capacities?

Sprinting horizontal force production capacities correspond to the maximal values of the force developed by an athlete onto the ground in the anteroposterior direction at any running velocities. These capacities decrease linearly when velocity increases, which is well described by the F-v relationship (see the graph 1.3 in the figure 1), [12] and its two extrema: *FH0* and *V0*, corresponding to the force production capacities at low and high velocities, respectively (figure 1) [12]. **Considering that sprinting horizontal force production capacities mostly depends on the lower limb and posterior chain muscles during sprinting action, [5,6] evaluation of *FH0* and *V0* can provide information about the lower limbs and posterior chain muscles function during the sprinting action.**

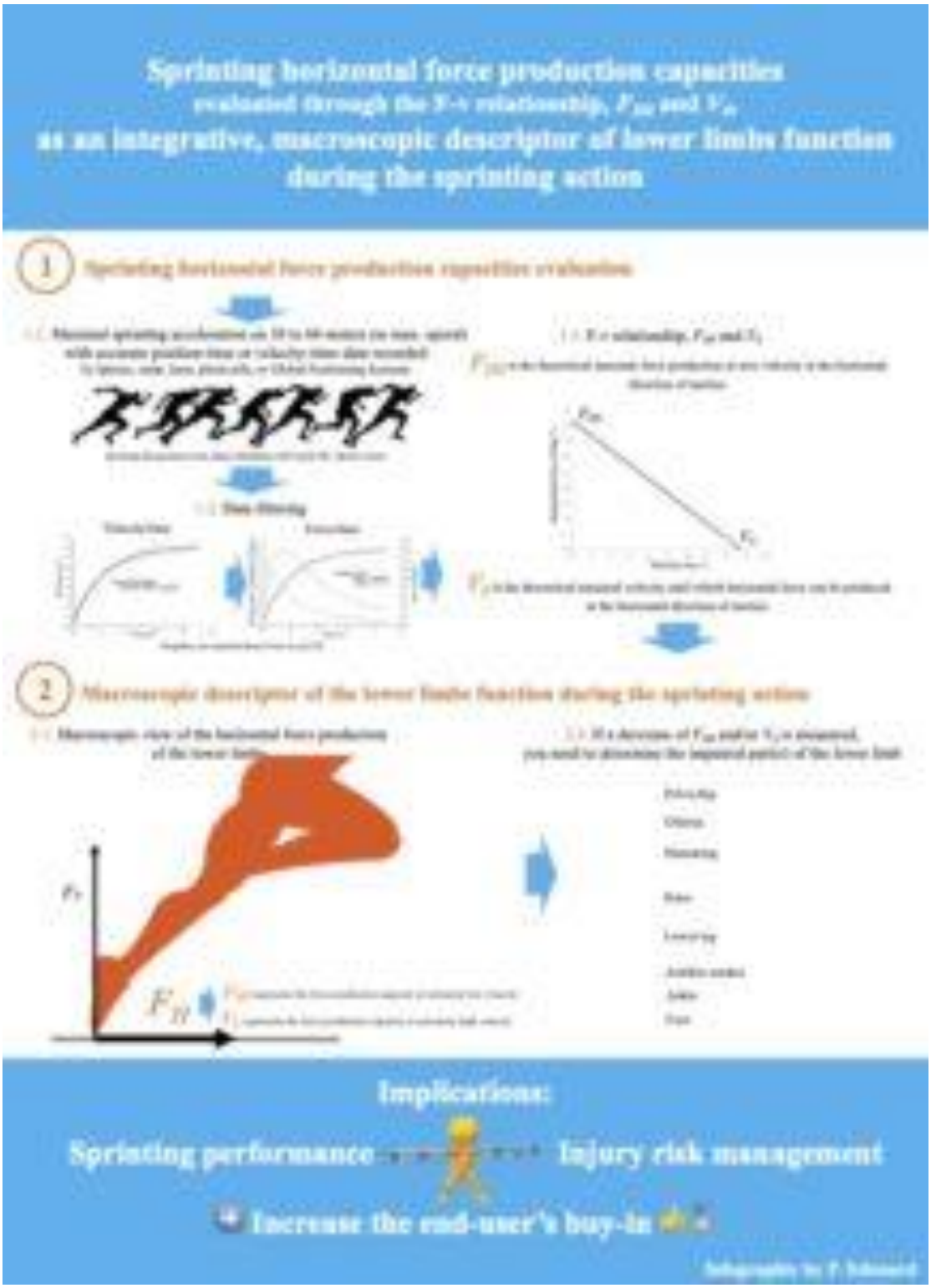

Figure 1: Illustration of the sprinting horizontal force production capacities evaluated by the F-v relationship, *FH0* and *V0* as descriptors of the lower limbs function during the sprinting action (*some graphics are inspired from Cross et al.[13]).*

How can we evaluate sprinting horizontal force production capacities in field practice?

Evaluation of the sprinting horizontal force production capacities requires a maximal sprint acceleration up to maximal running velocity (~30 to 60 metres) during which the position-time or velocity-time data are measured. Such measurements can be done using any technology that provides reliable data (e.g., radar, laser, 1080, photocells, or Global Positioning Systems, video camera [4,12]. Such maximal sprint accelerations are a common task for most trained athletes and can easily be included at the end of the warm-up session or in a specific sprinting training session. To obtain reliable and valid information, testing protocols, measurements and data analyses should be accurate and rigorously standardised (e.g., similar fatigue, training, time of day, warm-up conditions [12].

Why are sprinting horizontal force production capacities of interest for injury risk management?

Since sprinting horizontal force production capacities can provide information on the function of the lower limbs during sprinting, lower limb impairment can cause changes to these force capacities, especially impairment of the posterior chain muscles. When changes to the force capacities are not compensated by other parts of the kinetic chain during sprinting, sprinting horizontal force production capacities can decrease (i.e., decrease in *FH0* or *V0*). In other terms, the lower limbs are unable to produce the same levels of force in the horizontal direction over the sprinting entire acceleration. This can highlight a problem in the lower limb posterior chain. However, this is not specific to a single part of the chain. A *FH0* decrease has been reported after hamstring injury [7,8]. In addition, lower *FH0* were prospectively associated with higher risk of hamstring injury [9,11]. This could indicate that hamstring muscles could represent the weakest part of the kinetic chain. However, this is not always the case and decrease in sprinting horizontal force production capacities could be the consequence of other lower limb injuries, such as ankle sprain, calf muscle injury, knee pain, overtraining or fatigue. Therefore, sprinting horizontal force production capacities evaluation can help highlight an issue, with a more integrative, functional and performance-oriented approach, [9] justifying a subsequent deeper analysis. Such an evaluation can be done as a screening test [9–11] for primary prevention and/or as a return to sport test [7,8] for secondary prevention.

Increase the end-user's "buy-in" by killing two birds with one stone:

- Implementation of new strategies, measures or tools always represents a challenge. End-users now have lots of possibilities/opportunities to assist training and injury risk management, but often little time available. There is a need for efficiency.
- Since sprinting horizontal force production capacities evaluation is already used in routine for acceleration performance enhancement, no additional

measurement is required (from athletes or staff members) to generate information for lower limbs injury risk management.
- We therefore suggest an extension of the use of analysed data, with a two-way application opportunity: sprint acceleration training and injury risk management. This is a win-win performance-prevention approach.

Authors names & Affiliations:

Pascal Edouard 1,2, Jurdan Mendiguchia 3, Caroline Prince 4,5, Pedro Jimenez-Reyes 6, Kenny Guex 7,8, Johan Lahti 9, Pierre Samozino 4, Jean-Benoît Morin 1,10

1 Univ Lyon, UJM-Saint-Etienne, Inter-university Laboratory of Human Movement Biology, EA 7424, F-42023, Saint-Etienne, France

2 Department of Clinical and Exercise Physiology, Sports Medicine Unit, University Hospital of Saint-Etienne, Faculty of Medicine, Saint-Etienne, France

3 Department of Physical Therapy, ZENTRUM Rehab and Performance Center, Barañain, Spain

4 Univ Savoie Mont Blanc, Inter-university Laboratory of Human Movement Biology, EA 7424, F-73000 Chambéry, France

5 Physiotherapy department and motion analysis lab, Swiss Olympic Medical Center, La Tour Hospital, Meyrin, Switzerland

6 Centre for Sport Studies, Rey Juan Carlos University, Madrid, Spain

7 School of Health Sciences (HESAV), HES-SO University of Applied Sciences and Arts Western Switzerland, Lausanne, Switzerland

8 Swiss Athletics, Haus des Sports, Ittigen, Switzerland.

9 Université Côte d'Azur, LAMHESS, Nice, France

10 Sports Performance Research Institute New Zealand, School of Sport and Recreation, Auckland University of Technology, New Zealand

Correspondence to

Pascal Edouard, MD PhD, Department of Clinical and Exercise Physiology, Sports Medicine Unit, IRMIS, Campus Santé Innovations, University Hospital of Saint-Etienne, 42 055 Saint-Etienne cedex 2, France. Tel.: +33 674 574 691; Fax numbers: +33 477 127 229; E-mail: Pascal.Edouard@univ-st-etienne.fr

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